Nishal Stanislaus Silva, *PhD*

Computer Vision & ML Engineer | Industrial Inspection, IoT & Edge Deployment

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Professional Summary

Computer vision and machine learning engineer with 10+ years across industry and applied research, turning camera and multi-sensor streams into real-time decisions on deployed hardware. At MAS Holdings, shipped production computer vision for manufacturing at scale, including the camera-captured geometry alignment behind Promptly, now a commercial on-demand printing product, and care-label inspection that cut inspection time by 99.5%. Through a PhD and postdoc, took deep learning models from research prototype to deployment: 14 ms inference on a Raspberry Pi 4 at F1 0.76, 74x faster than the DTW baseline (JAES 2026), and multimodal sensor fusion of IMU and audio at F1 0.78 (IEEE IS2 2025). Rigorous experimental validation (benchmarks, ablations, open datasets), tested Python and C++, and 10+ peer-reviewed publications; now bringing this toolkit to self-driving perception.

Core Competencies

Computer Vision & Perception: OpenCV, feature matching, image registration and warping, segmentation, OCR, multi-camera arrays, defect detection, multimodal sensor fusion (IMU, audio, EEG, XR head/hand tracking)
ML Engineering: deep learning (LSTM/GRU/RNN, CNN), PyTorch, TensorFlow/Keras, synthetic data (VAE, diffusion), LoRA fine-tuning, multi-GPU training, benchmark and ablation design, large-scale evaluation
Systems & Deployment: real-time inference under latency budgets, C++17/Python inference on embedded hardware, ARM profiling, code efficiency, CI/CD, code review, unit testing

Technical Stack

Languages: Python, C++17, C, SQL, MATLAB, JavaScript/TypeScript, Shell
ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, SciPy, Librosa
Computer Vision: OpenCV, template & feature matching, OCR, image registration, industrial cameras, Cognex VisionPro, HALCON
Data, Cloud & MLOps: AWS (EC2, S3, ECS, RDS), ETL, Airflow, Docker, Jenkins, Git, Linux, pytest, CI/CD, Claude Code

Education

Ph.D - Information Engineering and Computer Science - *University of Trento, Italy* — *Jan 2025*
M.Sc - Telecommunication and Electronic Engineering - *Sheffield Hallam University, UK* — *Aug 2018*
B.Eng (Hons) - Electronic Engineering - *Sheffield Hallam University, UK* — *Mar 2014*

Professional Experience

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka* — *Jan 2015 - Jul 2020*

- Built end-to-end machine learning and computer-vision systems for manufacturing at scale, integrating camera, sensor and IoT streams into production workflows; operational efficiency improved by up to 300%.
- Promptly on-demand direct-to-garment printing: designed the geometry-alignment algorithm that maps camera-captured garment geometry to a per-side image warp, personally built the RIP integration, and consulted on the flipping mechanism; now a Twinery/MAS commercial product with facilities in the US, Mexico, France and Sri Lanka.

- Automated multilingual care-label QC with OpenCV template and feature matching and an explainable overlay, iterated from flatbed scanner to industrial camera to a conveyor with PLC reject; inspection time cut by 99.5%.
- Built loom-side fabric structural-defect detection with a microscopic camera array (overlapping fields of view across 60-inch rolls, 5-6 s alert budget); the pilot cut operator headcount by 50%.
- Ran C++/Python inference engines on embedded hardware and real-time ingestion of high-frequency IoT time-series across sites in South Asia, North America and Africa; introduced CI/CD and code review and mentored engineers.

Independent Research & Open-Source | *Self-directed, Toronto, Canada* *Jan 2026 - Present*

- Benchmarked 6 scoring methods across deterministic, probabilistic and deep-learning (Wav2Vec2) families over 100,000 scored trials on 10,000 sentences; best ROC-AUC 0.822 vs. 0.599 for the DTW baseline.
- Fine-tuning MusicGen-medium via LoRA with a custom audio-to-audio conditioning head, chosen after comparing 5 architectures; dual-GPU training (in progress). Built with AI-assisted workflows including Claude Code.

Postdoctoral Researcher | *University of Trento, Trento, Italy* *Jan 2025 - Dec 2025*

- Owned the full ML pipeline for streaming audio and IMU sensor data on the EU Horizon EIC Pathfinder project MUSMET (acquisition, feature engineering, training and evaluation, edge deployment, monitoring): 14 ms inference on Raspberry Pi 4, F1 0.76, 31.4% CPU, 74x faster than the DTW baseline (JAES 2026), validated with benchmark suites and ablations.
- Built a real-time multimodal synchronization pipeline fusing audio, BCI/EEG (g.tec) and mixed-reality head/hand tracking streams from 4 musicians simultaneously, with SAE Institute Barcelona.

Visiting Researcher | *McGill University (IDMIL / CIRMMT), Montreal, Canada* *May 2024 - Aug 2024*

- Multimodal sensor fusion on a self-powered smart guitar (BNO055 IMU at 100 Hz, audio interface, Raspberry Pi 4): an LSTM gesture detector on a 1 s accelerometer window fused with audio pattern detection via a hierarchical state machine; F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka* *Aug 2020 - Feb 2021*

- Deployed ML and statistical pipelines and SQL + Python scoring services for anomaly flagging on large structured enterprise datasets.

Selected Projects (full portfolio: *nishal.xyz/research*)

- **COVID-19 remote vitals monitoring:** webcams retrofitted to SpO2 and blood-pressure monitors with image processing feeding a remote dashboard, deployed across hospital wards (GMOA recognition).
- **Retail customer-behaviour vision pilot:** hot-spot detection of shopper behaviour for product placement.

Selected Publications (full list: *nishal.xyz/publications*)

- Silva, N. and Turchet, L. *Real-Time Audio Pattern Detection for Smart Musical Instruments*. Journal of the Audio Engineering Society, Vol. 74, No. 3, Mar. 2026.
- Silva, N., Wanderley, M. and Turchet, L. *Melody and Motion: Integrating Guitar Gestures with Musical Patterns for Extended Control in Live Performance and Metaverse Applications*. IEEE International Symposium on the Internet of Sounds, Oct. 2025.

Highlights & Recognition

- **Open datasets:** Four open-access datasets (DoMP, DoPP, DoDP, DoDP2), 9,800+ recordings from 70+ musicians, published on Zenodo as benchmarks for real-time pattern detection.
- **Recognition & service:** MIDI Innovation Awards 2023 finalist; reviewer, IEEE Access; Programme Committee, IEEE IS2 and IEEE I3DA; supervised 2 BSc theses.